Stochastic Discount Factor $\alpha\beta\gamma$

Structure, Estimation, Boundary and Long-term Factorization

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Integrated structure

1. Part I: **What's SDF**. stochastic discount factor language, no-arbitrage, HJ geometry, CCAPM, change of measure, and factor interpretation.
2. Part II: **SDF's Recovery and Factorization**. recovery, semigroups, path integrals, Hansen–Scheinkman, and operator-based long-run factorization.
3. Part III: **Three Related Papers**. on entropy recovery, SDF moments, and principal portfolios.

Part I

Asset Pricing Through the Lens of the SDF

Roadmap: one object, four tasks

Central pricing object

$$p_t = \mathbb{E}_t[m_{t+1}x_{t+1}], \qquad 1 = \mathbb{E}_t[m_{t+1}R_{t+1}], \qquad P_t = \frac{1}{\Lambda_t}\mathbb{E}_t\left[\int_t^T \Lambda_u D_u \, \mathrm{d}u + \Lambda_T X_T\right].$$

1. **Why does an admissible SDF exist?**
   - Arrow prices, no-arbitrage, FTAP, positivity.
2. **What do traded returns reveal about it?**
   - frontier geometry, maximal Sharpe ratio, HJ bound, traded projection.
3. **What economic structure can be imposed on it?**
   - portfolio choice, Euler equations, CCAPM-type restrictions.
4. **How do we reweight or estimate it?**
   - change of measure, factor approximations, joint tests, SDF estimation.

The pricing kernel as the unifying language

Definition (Stochastic discount factor)

A strictly positive random variable m_{t+1} is an SDF for one-period payoffs if every asset payoff x_{t+1} with time-t price p_t satisfies

$$p_t = \mathbb{E}_t^{\mathbb{P}}[m_{t+1}x_{t+1}], \qquad 1 = \mathbb{E}_t^{\mathbb{P}}[m_{t+1}R_{t+1}].$$

In continuous time it is often more natural to embed one-period pricing in a strictly positive *state-price density* process $\{\Lambda_t\}_{t\geq 0}$:

$$P_t = \frac{1}{\Lambda_t}\mathbb{E}_t^{\mathbb{P}}\left[\int_t^T \Lambda_u D_u \, \mathrm{d}u + \Lambda_T X_T\right], \qquad m_{t,t+\Delta} \approx \frac{\Lambda_{t+\Delta}}{\Lambda_t}.$$

What this buys us

No-arbitrage explains *existence*; portfolio geometry reveals *restrictions*; preference-based models impose *structure*; change of measure and empirical methods deliver *reweighting and estimation*.

Formal definitions and notation: Cochrane, *Asset Pricing*, Chs. 1–5; Duffie, *Dynamic Asset Pricing Theory*, Chs. 1–2.

Immediate implications of the pricing equation

For any excess return R^e_{t+1},

$$0 = \mathbb{E}_t[m_{t+1}R^e_{t+1}] = \mathbb{E}_t[m_{t+1}]\mathbb{E}_t[R^e_{t+1}] + \mathrm{Cov}_t(m_{t+1}, R^e_{t+1}),$$

so that

$$\mathbb{E}_t[R^e_{t+1}] = -\frac{\mathrm{Cov}_t(m_{t+1}, R^e_{t+1})}{\mathbb{E}_t[m_{t+1}]}, \qquad R^f_t = \frac{1}{\mathbb{E}_t[m_{t+1}]}.$$

Separate the two margins immediately

$$\text{level of discounting} \iff \mathbb{E}_t[m_{t+1}],$$
$$\text{risk premia} \iff -\mathrm{Cov}_t(m_{t+1}, R^e_{t+1}).$$

Standard derivations: Cochrane, Chs. 1 and 5.

No-arbitrage and the First Fundamental Theorem

Definition (Arbitrage, one-period finite-asset formulation)

An arbitrage is a portfolio with zero or negative initial cost, nonnegative payoff in every state, and strictly positive payoff with positive probability.

Theorem (First Fundamental Theorem of Asset Pricing, finite-state version)

Suppose the asset payoff matrix is finite dimensional. Then absence of arbitrage is equivalent to the existence of a strictly positive linear pricing functional. Equivalently, there exists a strictly positive SDF m such that

$$p_t = \mathbb{E}_t[m_{t+1}x_{t+1}], \qquad \textit{equivalently} \qquad 1 = \mathbb{E}_t[m_{t+1}R_{t+1}]$$

for every traded payoff or gross return.

In state-price notation, if $q(s)$ is the Arrow-Debreu price of one unit delivered in state s, then

$$p = \sum_s q(s)x(s), \qquad m(s) = \frac{q(s)}{\mathbb{P}(s)}, \qquad q(s) \geq 0.$$

Formal statements and extensions: Duffie, Chs. 1 and 6; Cochrane, Ch. 4.

Mean-variance frontier and the maximal Sharpe ratio

Let $R \in \mathbb{R}^N$ denote the vector of asset gross returns, $\mu = \mathbb{E}[R]$, and $\Sigma = \mathrm{Var}(R)$. For portfolio weights w satisfying $w'\mathbf{1} = 1$,

$$R_p = w'R, \qquad \mathbb{E}[R_p] = w'\mu, \qquad \mathrm{Var}(R_p) = w'\Sigma w.$$

The mean-variance problem is

$$\min_w \; w'\Sigma w \qquad \text{s.t.} \qquad w'\mathbf{1} = 1, \; w'\mu = \bar{\mu}.$$

If a risk-free asset exists, the tangency portfolio maximizes

$$\mathrm{SR}(R_p) = \frac{\mathbb{E}[R_p] - R_f}{\sigma(R_p)}.$$

Two-fund separation, in words

Once a risk-free asset is available, every mean-variance efficient portfolio can be generated by combining only two funds: the risk-free asset and the tangency portfolio. Hence the efficient set is summarized by one distinguished risky direction in traded return space.

Frontier geometry and its connection to discount factors: Cochrane, Chs. 5 and 6.

Hansen-Jagannathan bound as a restriction on admissible SDFs

Hansen-Jagannathan bound

If m prices all traded excess returns, then for every excess return R^e,

$$\frac{|\mathbb{E}[R^e]|}{\sigma(R^e)} \leq \frac{\sigma(m)}{\mathbb{E}[m]}.$$

Taking the supremum over all traded excess returns gives

$$\sup_{R^e} \frac{|\mathbb{E}[R^e]|}{\sigma(R^e)} \leq \frac{\sigma(m)}{\mathbb{E}[m]}.$$

One-line proof sketch

From $0 = \mathbb{E}[mR^e]$ and Cauchy–Schwarz,

$$|\mathbb{E}[m]\mathbb{E}[R^e]| = |\mathrm{Cov}(m, R^e)| \leq \sigma(m)\sigma(R^e).$$

- The left-hand side is the maximal Sharpe ratio attainable in the traded return space.
- The right-hand side is a lower bound on the coefficient of variation of any admissible m.

Hansen and Jagannathan (1991, *JPE*); Cochrane, Ch. 5.

The class and tradedable projection of the SDF

Let

$$\mathcal{M}_R := \{a - b'R : a \in \mathbb{R},\ b \in \mathbb{R}^N\}$$

be the affine span generated by traded gross returns. Among all admissible SDFs satisfying

$$\mathbb{E}[mR] = \mathbf{1},$$

define the minimum-second-moment solution by

$$m^\star := \arg\min_{\tilde{m}} \ \mathbb{E}[\tilde{m}^2] \qquad \text{s.t.} \qquad \mathbb{E}[\tilde{m}R] = \mathbf{1}.$$

Projection and orthogonality

The optimizer satisfies $m^\star \in \mathcal{M}_R$, and for any other admissible SDF m,

$$m = m^\star + \varepsilon, \qquad \mathbb{E}[\varepsilon R] = 0.$$

Equivalently,

$$m^\star = \mathrm{Proj}_{\mathcal{M}_R}(m)$$

with respect to the L^2 inner product induced by $\mathbb{E}[\cdot\,,\cdot]$.

Hansen and Richard (1987); Hansen and Jagannathan (1991); Martin (2017, 2019).

The general one-period portfolio problem

Consider a representative investor at date t who chooses current consumption c_t and portfolio holdings θ over contingent payoffs x_{t+1}:

$$\max_{c_t,\theta} \; u(c_t) + \beta \mathbb{E}_t\left[u(c_{t+1})\right]$$

subject to

$$c_t + p_t'\theta = e_t, \qquad c_{t+1}(s) = e_{t+1}(s) + x_{t+1}(s)'\theta.$$

Proposition (General one-period first-order condition)

If the optimum is interior and differentiable, then for every traded payoff x^i_{t+1},

$$p^i_t u'(c_t) = \beta \mathbb{E}_t\left[u'(c_{t+1}) x^i_{t+1}\right].$$

Equivalently,

$$p^i_t = \mathbb{E}_t\left[m_{t+1} x^i_{t+1}\right], \qquad m_{t+1} = \beta \frac{u'(c_{t+1})}{u'(c_t)}.$$

Interpretation

At an optimized allocation, there is no first-order gain from an infinitesimal transfer across *time* or across *states*. The same marginal utility price must support every small reallocation.

One-period optimality and Euler conditions: Cochrane, Ch. 1; Duffie, Ch. 9.

Euler equations, Arrow prices, and the IMRS

If one writes the budget set in terms of Arrow securities with state prices $q_t(s)$, the same first-order logic gives

$$q_t(s)u'(c_t) = \beta \mathbb{P}_t(s) u'(c_{t+1}(s)).$$

Hence

$$q_t(s) = \mathbb{P}_t(s)\, \beta \frac{u'(c_{t+1}(s))}{u'(c_t)} = \mathbb{P}_t(s)\, m_{t+1}(s).$$

Therefore the SDF is simultaneously

- the intertemporal marginal rate of substitution,
- the ratio of state prices to physical probabilities,
- and the object that makes all first-order conditions consistent.

Arrow-security interpretation: Cochrane, Chs. 1 and 4; Duffie, Ch. 1.

CCAPM as a structured model for the SDF

With constant relative risk aversion utility,

$$u(C) = \frac{C^{1-\gamma}}{1-\gamma}, \qquad u'(C) = C^{-\gamma},$$

so the SDF becomes

$$m_{t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma}.$$

The Euler equation is therefore

$$1 = \mathbb{E}_t \left[\beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} R^i_{t+1} \right].$$

The implied risk-premium relation is

$$\mathbb{E}_t[R^{e,i}_{t+1}] = -\frac{\mathrm{Cov}_t\left(\beta(C_{t+1}/C_t)^{-\gamma}, R^{e,i}_{t+1}\right)}{\mathbb{E}_t[\beta(C_{t+1}/C_t)^{-\gamma}]}.$$

Comment

CCAPM is a concrete proposal for the relevant state variable: aggregate consumption growth is assumed to span the economically relevant part of m.

Lucas-style consumption-based pricing kernel: Cochrane, Ch. 1; Duffie, Ch. 9.

When the basic CCAPM fails: anomalies as diagnostics of misspecified m

- **Equity premium puzzle**: observed average excess returns are too large relative to the smoothness of aggregate consumption growth.
- **Risk-free rate puzzle**: simple CRRA calibrations struggle to match both the level of risk premia and the level of the safe rate.
- **Cross-sectional anomalies**: size, value, momentum, profitability, and investment patterns are not well captured by a basic consumption beta alone.

Model class	How it enriches the SDF
Epstein-Zin recursive utility	decouples risk aversion from the elasticity of intertemporal substitution
Habit formation	makes marginal utility highly state dependent in bad times
Long-run risk / rare disasters	loads m on persistent growth risk and tail risk
Heterogeneous-agent models	makes the aggregate kernel richer than representative-agent consumption growth

Comment

The force of the anomalies is not that asset pricing theory fails. It is that a particular functional form for m is too rigid.

For puzzles and solution branches: Cochrane, Chs. 20–21; Campbell and Cochrane (1999); Bansal and Yaron (2004); Epstein and Zin (1989).

From one-period SDFs to state-price density processes

The discrete-time kernel m_{t+1} prices one-period payoffs. In continuous time it is more natural to work with a strictly positive process Λ_t such that short-horizon SDFs are generated by ratios of that process:

$$m_{t,t+\Delta} \approx \frac{\Lambda_{t+\Delta}}{\Lambda_t}.$$

With a money-market account $B_t = e^{\int_0^t r_s \, \mathrm{d}s}$, write

$$\Lambda_t = \frac{Z_t}{B_t}, \qquad Z_t := \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\bigg|_{\mathcal{F}_t}.$$

Thus the state-price density separates naturally into

$$\text{time value of money} \iff B_t^{-1},$$
$$\text{probability reweighting} \iff Z_t.$$

What if without a given risk-neutral probability?

Once the kernel is written as a state-price density process, Radon–Nikodym derivatives, equivalent martingale measures, Girsanov, and Feynman–Kac become different ways of describing the same pricing operator. In other words, it's a process to identify a numeraire.

Continuous-time pricing kernel notation: Duffie, Chs. 6 and 10.

From the SDF to a risk-neutral measure

Start from the discrete-time pricing equation

$$p_t = \mathbb{E}_t^{\mathbb{P}}[m_{t+1} x_{t+1}].$$

Define the one-period likelihood ratio

$$\Lambda_{t+1} := \frac{m_{t+1}}{\mathbb{E}_t[m_{t+1}]} = R_t^f m_{t+1},$$

so that $\mathbb{E}_t^{\mathbb{P}}[\Lambda_{t+1}] = 1$. Then define a new measure $\mathbb{Q}$ by

$$\left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_{t+1}} = \Lambda_{t+1}.$$

Therefore

$$p_t = \frac{1}{R_t^f} \mathbb{E}_t^{\mathbb{Q}}[x_{t+1}].$$

Change of numeraire and equivalent martingale measures: Duffie, Chs. 6 and 10.

Equivalent martingale measures and official FTAP language

Definition (Equivalent martingale measure)

A probability measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ is an equivalent martingale measure (EMM) for a chosen numeraire if discounted asset prices are $\mathbb{Q}$-martingales.

Theorem (First FTAP, martingale version — informal textbook statement)

Under suitable market and admissibility conditions, absence of arbitrage is equivalent to the existence of an equivalent martingale measure.

For a money-market account B_t, the pricing formula becomes

$$P_t = B_t \, \mathbb{E}_t^{\mathbb{Q}} \left[\frac{X_T}{B_T} \right].$$

The process representation is

$$\Lambda_t = \frac{Z_t}{B_t}, \qquad Z_t := \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\Big|_{\mathcal{F}_t}.$$

Duffie, Chs. 6 and 10; Harrison and Kreps (1979); Harrison and Pliska (1981).

Continuous-time setup and Girsanov's theorem

Suppose under $\mathbb{P}$ the state vector X_t satisfies

$$\mathrm{d}X_t = \mu^{\mathbb{P}}(X_t, t)\ \mathrm{d}t + \sigma(X_t, t)\ \mathrm{d}W_t^{\mathbb{P}},$$

where $W_t^{\mathbb{P}}$ is a Brownian motion. Let the market price of risk process be λ_t. Define the density process

$$Z_t = \exp\left(-\int_0^t \lambda_s' \,\mathrm{d}W_s^{\mathbb{P}} - \frac{1}{2}\int_0^t \|\lambda_s\|^2 \,\mathrm{d}s\right).$$

Theorem (Girsanov)

If Z_t is a true martingale, then the measure $\mathbb{Q}$ defined by

$$\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\Big|_{\mathcal{F}_t} = Z_t$$

changes the Brownian motion to

$$\mathrm{d}W_t^{\mathbb{Q}} = \mathrm{d}W_t^{\mathbb{P}} + \lambda_t\,\mathrm{d}t,$$

so that under $\mathbb{Q}$ the state dynamics become

$$\mathrm{d}X_t = \mu^{\mathbb{Q}}(X_t, t)\ \mathrm{d}t + \sigma(X_t, t)\ \mathrm{d}W_t^{\mathbb{Q}},$$

with the diffusion term unchanged and the drift adjusted by risk prices.

Continuous-time change of measure: Duffie, Ch. 6; standard probability texts on Girsanov.

Feynman-Kac as the bridge from pricing operators to PDEs

Let the claim price at time t be $v(t, X_t)$, paying flow dividend $d(X_t, t)$ and terminal payoff $g(X_T)$. Under the risk-neutral measure,

$$v(t,x) = \mathbb{E}^{\mathbb{Q}}_{t,x}\left[\int_t^T e^{-\int_t^u r_s \,\mathrm{d}s} d(X_u, u)\,\mathrm{d}u + e^{-\int_t^T r_s \,\mathrm{d}s} g(X_T)\right].$$

This is the final step of the change-of-measure module: Radon–Nikodym derivatives reweight states, Girsanov changes the drift under the new measure, and Feynman–Kac turns the resulting pricing operator into a PDE representation.

Theorem (Feynman-Kac, diffusion form — textbook statement)

If X_t is a Markov diffusion under $\mathbb{Q}$ with generator $\mathcal{L}^{\mathbb{Q}}$ and v is sufficiently regular, then v solves

$$\partial_t v + \mathcal{L}^{\mathbb{Q}} v - rv + d = 0, \qquad v(T,x) = g(x).$$

Conversely, the sufficiently regular PDE solution admits the conditional-expectation representation above.

Feynman-Kac representation in finance: Duffie, Ch. 10.

EMH and the joint hypothesis problem

Definition (Fama (1970), textbook statement of market efficiency)

A market is efficient with respect to an information set if security prices "fully reflect" that information set, so that price changes occur only when new information arrives.

Joint hypothesis problem

Every empirical test of market efficiency is jointly a test of

$$\text{(i) informational efficiency} \;+\; \text{(ii) a maintained asset-pricing model.}$$

In SDF language the maintained model is a maintained restriction on the admissible kernel,

$$\mathbb{E}[m_{t+1} R^{e,i}_{t+1}] = 0, \qquad i = 1, \ldots, N.$$

Hence rejecting zero-alpha restrictions may reflect either market inefficiency or a misspecified SDF.

Fama (1970, 1991) for EMH; Cochrane, Ch. 12 for the asset-pricing and joint-test interpretation.

Why linear factor models appear, formally

Suppose we approximate the pricing kernel by a linear factor SDF,

$$m_{t+1} = a_t - b_t' f_{t+1}.$$

Then the pricing conditions become

$$1 = \mathbb{E}_t[(a_t - b_t' f_{t+1}) R^i_{t+1}],$$

which can be rewritten under standard assumptions as a beta representation,

$$\mathbb{E}[R^{e,i}_{t+1}] = \beta_i' \lambda, \qquad \beta_i = \frac{\mathrm{Cov}(f_{t+1}, R^i_{t+1})}{\mathrm{Var}(f_{t+1})}.$$

Why linearity persists

1. it is the simplest projection of the traded component of m;
2. it yields testable moment restrictions and beta-pricing relations;
3. it provides a direct bridge from characteristics to factors to expected returns.

Linear factor SDFs and beta representations: Cochrane, Chs. 6, 9, and 12.

Characteristics, factors, and Cochrane's three questions

Cochrane's "factor zoo" challenge can be read as a challenge about the dimension and structure of the SDF. The three questions are:

1. Which characteristics really provide independent information about average returns?
2. Which are subsumed by others?
3. Does each new anomaly variable also correspond to a new factor formed on those same anomalies, and how many of these factors are really important?

SDF interpretation

A new anomaly matters only if it reveals a genuinely new direction in priced payoffs, or a genuinely new factor needed to approximate m.

Cochrane (2011, *Discount Rates*); Kelly, Pruitt, and Su (2019, *JFE*) provide an explicit covariance-based response.

Existing factor models and their relation to the SDF

Model class	What is specified	Relation to m
CAPM / ICAPM	benchmark return factors	low-dimensional traded approximation to m
Fama-French style models	long-short factor portfolios formed on characteristics	empirical spanning approximation for the traded kernel
Consumption-based models	macro state variables and preferences	structural specification for the economic kernel
Latent factor models	hidden statistical factors	low-rank approximation to the priced return space
Characteristic models	average returns projected on firm characteristics	reduced-form proxy for exposures or conditional expected returns

The common discipline is always the same:

does the proposed factor structure capture the part of m that prices the assets under study?

If the answer is yes, the factor model is a useful approximation; if the answer is no, the model is merely relabeling already priced information or missing an important direction in the SDF.

Cochrane, Chs. 12 and 13; the factor-zoo discussion is sharpened in Cochrane (2011).

A unified map of SDF estimation

Data	Restriction on m	Typical methods
Returns only	moment conditions $\mathbb{E}[mR] = 1$	GMM, cross-sectional beta pricing, HJ-distance minimization
Macro quantities	preference-implied functional form	CCAPM, recursive utility, habit, long-run risk
Derivative prices	state-price information under $\mathbb{Q}$ plus links back to $\mathbb{P}$	risk-neutral density estimation, recovery, affine term structure methods
Latent factor methods	low-rank or flexible approximation to priced covariance structure	PCA, IPCA, sieve methods, machine-learning SDF estimators

Unifying view

Different empirical literatures often look disconnected because they use different observables. From the SDF viewpoint they are all identification strategies for the same latent pricing object.

Takeaway: SDF Matters!

- The most coherent way to teach asset pricing is to teach the stochastic discount factor.
- No-arbitrage ensures that a strictly positive pricing kernel exists.
- Portfolio geometry reveals what traded returns imply about the volatility and projection of that kernel.
- Euler equations impose economic structure on the same object.
- Change of measure rewrites the same object as a density process between $\mathbb{P}$ and $\mathbb{Q}$.
- Factor models, characteristic models, and modern estimators are all attempts to identify or approximate the relevant component of m.

[Definition Problem]

Selection — or asset pricing itself — is intrinsically an optimization problem.

[Three Key Constraints]

- Any Arrow-Debreu security (or contingent claim) must carry a non-negative price.
- Probability and portfolio weights are both normalized objects with components summing to one.
- For an optimized allocation, no first-order gain remains from an infinitesimal shift across time or across states.

Closing thesis

Asset pricing is, at its core, a structured identification problem for an unobserved pricing object.

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Part II

Recovery, Semigroup, and Long-Run Factorization

Ross recovery: what the paper tries to recover

Central object

Given Arrow state prices or option-implied risk-neutral information, Ross asks whether one can recover the *natural* transition matrix F, the discount rate δ, and the pricing kernel ϕ rather than only the martingale measure.

- Recovery targets transition probabilities under P, not primitive preferences themselves.
- The economic attraction is decomposition: separate discounting, state dynamics, and risk adjustment.
- The mathematical attraction is that the problem becomes an eigenvalue/eigenfunction problem for a positive pricing operator.

Why this matters in the lecture

This is the cleanest place where SDF language, operator factorization, option data, and EMH-style empirical restrictions all touch the same object.

Assumptions that make recovery work

Core assumptions enabling uniqueness

- Time-homogeneous environment.
- Markovian state dynamics, in discrete time or via a continuous-time generator.
- Transition independence / separability: the intertemporal valuation part can be written independently of the physical state transition kernel.
- Positive, irreducible pricing matrix so that Perron–Frobenius theory applies.

$$\text{state-price matrix} = \text{discount factor} \times \text{relative state values} \times \text{natural transition probabilities}$$

Intuition

Once the pricing kernel is restricted to move through a one-dimensional positive eigenstructure, observed Arrow prices can pin down the hidden transition law.

Basic conditional pricing setup

Suppose states follow a time-homogeneous Markov process. Let $p(\theta_i, \theta_j)$ denote the one-step state price from current state θ_i to next state θ_j, and let $f(\theta_i, \theta_j)$ denote the corresponding natural transition probability.

Transition-independence factorization

$$\phi(\theta_i, \theta_j) = \frac{p(\theta_i, \theta_j)}{f(\theta_i, \theta_j)} = \delta \, \frac{h(\theta_j)}{h(\theta_i)}.$$

Hence

$$p(\theta_i, \theta_j) = \delta \, \frac{h(\theta_j)}{h(\theta_i)} f(\theta_i, \theta_j).$$

- Conditional price = conditional pricing kernel $\times$ natural transition probability.
- h measures relative continuation value across states.
- δ captures pure time discounting.

Ross recovery theorem

Theorem 1: Recovery theorem

If there is no arbitrage, the pricing matrix is irreducible, and pricing is generated by a transition-independent kernel, then there exists a unique positive solution for the natural transition matrix F, the discount rate δ, and the pricing kernel ϕ compatible with the observed state prices.

Related statements emphasized in the original slides

- If the riskless rate is state independent, the unique natural density coincides with the martingale density.
- The subjective discount rate δ is bounded above by the largest interest factor.

Interpretation

Uniqueness does not come from option prices alone. It comes from option prices *plus* the transition-independence restriction.

Consumption-based interpretation

Consider a representative agent with additively separable preferences and constant discount factor δ:

$$\max_{\{c(\theta_i),c(\theta)\}} \left\{ U(c(\theta_i)) + \delta \int U(c(\theta)) f(\theta_i, \theta)\, d\theta \right\}$$

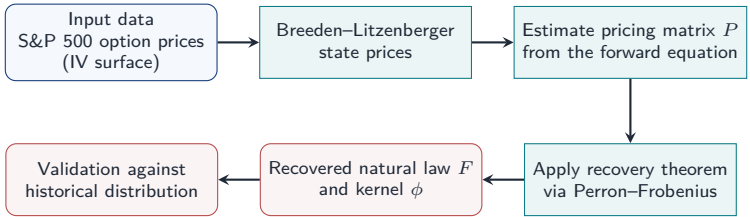
subject to the one-step budget constraint.

Kernel under the consumption interpretation

$$\phi(\theta_i, \theta_j) = \delta \frac{h(\theta_j)}{h(\theta_i)} = \delta \frac{U'(c(\theta_j))}{U'(c(\theta_i))} \quad \text{under transition independence.}$$

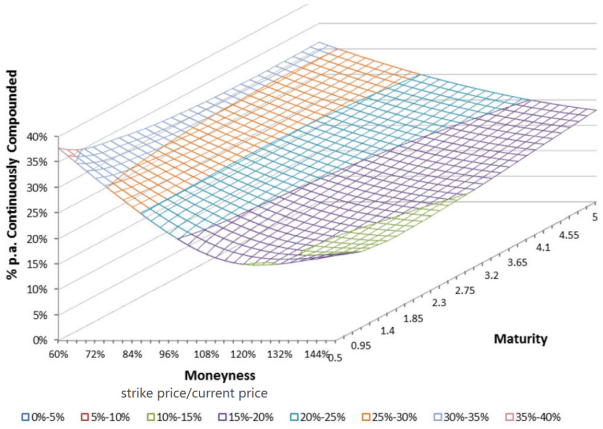
- The natural density stochastically dominates the risk-neutral density in the consumption result emphasized by Ross.
- This links recovery to a familiar SDF ratio of marginal utilities, but the theorem itself is broader than a specific parametric utility choice.

Option-based implementation: empirical workflow



- Step 1 extracts Arrow state prices from option data.
- Step 2 imposes time homogeneity to estimate a full transition structure from the observed term structure.
- Step 3 recovers F and ϕ through the dominant eigenvalue/eigenvector pair.

From implied volatility to state prices



- Volatility Smirk
- Term Structure

The implied-volatility smirk reveals strong downside asymmetry. The option surface is first turned into risk-neutral state prices; recovery then asks whether this forward-looking object can be decomposed into natural probabilities and a pricing kernel.

Estimating the pricing matrix from option data

Panel A: State Prices on April 27, 2011

Return\Tenor	0.25	0.5	0.75	1	1.25	1.5	1.75	2	2.25	2.5	2.75	3
−36%	0.005	0.023	0.038	0.050	0.058	0.064	0.068	0.071	0.073	0.075	0.076	0.076
−30%	0.007	0.019	0.026	0.030	0.032	0.034	0.034	0.035	0.035	0.035	0.034	0.034
−24%	0.018	0.041	0.046	0.050	0.051	0.052	0.051	0.050	0.050	0.049	0.048	0.046
−16%	0.045	0.064	0.073	0.073	0.072	0.070	0.068	0.066	0.064	0.061	0.058	0.056
−9%	0.164	0.156	0.142	0.128	0.118	0.109	0.102	0.096	0.091	0.085	0.081	0.076
0%	0.478	0.302	0.234	0.198	0.173	0.155	0.141	0.129	0.120	0.111	0.103	0.096
9%	0.276	0.316	0.278	0.245	0.219	0.198	0.180	0.164	0.151	0.140	0.130	0.120
20%	0.007	0.070	0.129	0.155	0.166	0.167	0.164	0.158	0.152	0.145	0.137	0.130
31%	0.000	0.002	0.016	0.036	0.055	0.072	0.085	0.094	0.100	0.103	0.105	0.105
43%	0.000	0.000	0.001	0.004	0.009	0.017	0.026	0.036	0.045	0.053	0.061	0.067
57%	0.000	0.000	0.000	0.000	0.000	0.000	0.001	0.001	0.002	0.002	0.003	0.003
Discount Factor	1.000	0.993	0.983	0.969	0.953	0.938	0.920	0.900	0.883	0.859	0.836	0.809

Panel B: The State Price Transition Matrix (P)

	Sigmas	−5	−4	−3	−2	−1	0	1	2	3	4	5
Sigmas	$S_0 \backslash S_T$	0.64	0.70	0.76	0.84	0.91	1	1.09	1.20	1.31	1.43	1.57
−5	0.64	0.671	0.241	0.053	0.005	0.001	0.001	0.001	0.001	0.001	0.000	0.000
−4	0.70	0.280	0.396	0.245	0.054	0.004	0.000	0.000	0.000	0.000	0.000	0.000
−3	0.76	0.049	0.224	0.394	0.248	0.056	0.004	0.000	0.000	0.000	0.000	0.000

(Continued)

Panel A (State Prices)
Panel B (Estimated Transition Matrix P):

- Direct input for the Recovery Theorem.
- Row sums represent discount factors (not equal to 1).

Starting from Breeden–Litzenberger state prices across strikes and maturities, the empirical object is the pricing matrix. Panel A extracts state prices; Panel B reorganizes them into the matrix input for recovery.

Recovered natural probabilities and the pricing kernel

Table 3: Recovered Natural Probabilities & Kernel

Implied volatility $\xrightarrow{BS}$ risk neutral P* $\xrightarrow{SDF}$ state price $\xrightarrow[matrixP]{recovery}$ $p \times \phi$

The Natural Probability Transition Matrix (F)

	Sigmas	−5	−4	−3	−2	−1	0	1	2	3	4	5
Sigmas	$S_0 \backslash S_T$	−36%	−30%	−24%	−16%	−9%	0%	9%	20%	31%	43%	57%
−5	−36%	0.670	0.253	0.061	0.006	0.002	0.001	0.002	0.001	0.002	0.002	0.000
−4	−30%	0.266	0.395	0.267	0.066	0.005	0.000	0.001	0.000	0.000	0.000	0.000
−3	−24%	0.043	0.205	0.393	0.278	0.073	0.007	0.000	0.000	0.000	0.000	0.000
−2	−16%	0.004	0.035	0.193	0.390	0.290	0.081	0.006	0.000	0.000	0.000	0.000
−1	−9%	0.004	0.005	0.031	0.181	0.385	0.309	0.080	0.005	0.000	0.000	0.000
0	0%	0.003	0.004	0.011	0.031	0.132	0.477	0.333	0.010	0.000	0.000	0.000
1	9%	0.000	0.000	0.000	0.002	0.027	0.169	0.381	0.314	0.095	0.011	0.000
2	20%	0.000	0.000	0.000	0.001	0.003	0.028	0.163	0.373	0.318	0.102	0.013
3	31%	0.000	0.000	0.000	0.001	0.001	0.003	0.025	0.148	0.361	0.330	0.130
4	43%	0.000	0.000	0.000	0.000	0.000	0.000	0.001	0.019	0.131	0.347	0.501
5	57%	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.014	0.113	0.873
Kernel, ϕ =		1.86	1.77	1.62	1.44	1.24	1	0.83	0.66	0.50	0.35	0.22

- **Eigenvalue (δ)**: Time discount factor
- **Eigenvector (z)**: Related to the Pricing Kernel(risk aversion)
- **Matrix F**: The recovered Natural Probability Matrix

$$f_{ii} = \frac{1}{\delta} \times \frac{z_j}{z_i} \times p_{ii}$$

If $Pz = \delta z$, then recovery rewrites observed Arrow prices as $f_{ij} = \frac{1}{\delta} \frac{z_j}{z_i} p_{ij}$, so the dominant eigenpair converts observed state prices into a candidate natural transition law.

Recovered vs. historical distributions

Table 4: Recovered vs. Historical Distributions

Implied volatility $\xrightarrow{BS}$ risk neutral P* $\xrightarrow{SDF}$ state price $\xrightarrow[P]{recovery}$ $p \times \phi$

Panel A: The Recovered Marginal Distributions

Return\Tenor	0.25	0.5	0.75	1	1.25	1.5	1.75	2	2.25	2.5	2.75	3
−36%	0.003	0.012	0.020	0.026	0.030	0.032	0.034	0.036	0.037	0.038	0.038	0.039
−30%	0.004	0.010	0.014	0.016	0.017	0.018	0.018	0.018	0.018	0.018	0.018	0.018
−24%	0.011	0.025	0.028	0.030	0.030	0.030	0.030	0.029	0.029	0.028	0.028	0.027
−16%	0.031	0.044	0.049	0.049	0.047	0.046	0.044	0.043	0.041	0.039	0.038	0.038
−9%	0.132	0.124	0.111	0.099	0.090	0.083	0.077	0.072	0.068	0.065	0.062	0.059
0%	0.477	0.299	0.228	0.190	0.165	0.146	0.132	0.121	0.112	0.104	0.098	0.092
9%	0.333	0.377	0.327	0.285	0.252	0.225	0.203	0.185	0.171	0.159	0.148	0.140
20%	0.010	0.105	0.190	0.226	0.239	0.238	0.232	0.224	0.215	0.206	0.197	0.189
31%	0.000	0.005	0.031	0.068	0.104	0.134	0.157	0.173	0.184	0.192	0.197	0.200
43%	0.000	0.000	0.002	0.010	0.025	0.045	0.070	0.094	0.118	0.141	0.163	0.182
57%	0.000	0.000	0.000	0.000	0.001	0.002	0.003	0.005	0.007	0.009	0.012	0.015

Panel B: Recovered Summary Statistics (Annualized)

	0.25	0.5	0.75	1	1.25	1.5	1.75	2	2.25	2.5	2.75	3
Mean	0.051	0.055	0.06	0.062	0.063	0.063	0.063	0.062	0.061	0.06	0.058	0.057
Sigma	0.117	0.14	0.147	0.15	0.152	0.153	0.154	0.154	0.153	0.152	0.151	0.149
Risk-free	0.005	0.002	0.01	0.018	0.026	0.034	0.041	0.048	0.054	0.061	0.068	0.074
$E-r$	0.047	0.053	0.05	0.043	0.036	0.03	0.022	0.015	0.007	−0.001	−0.009	−0.017
Sharpe	0.399	0.376	0.34	0.287	0.239	0.193	0.146	0.096	0.044	−0.008	−0.061	−0.115
ATM vol.	0.145	0.167	0.177	0.182	0.185	0.188	0.191	0.193	0.196	0.198	0.201	0.203

(Continued)

1. Fat Tail Risk
2. Recovered Mean ($\approx 6\%$) $<$ Historical Mean ($\approx 10\%$).

Recovered vs. bootstrapped histogram

Figure 2: Recovered vs.Bootstrapped histogram

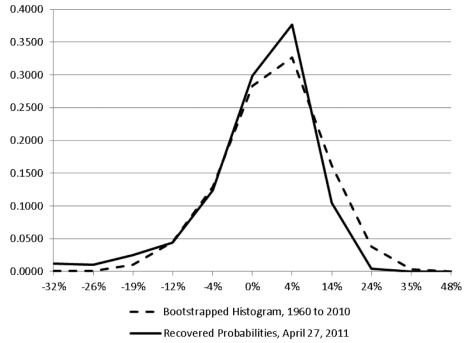


Figure 2. The recovered and the bootstrapped natural densities.

- underlying price $\rightarrow$ statiscal bootstrapped result
- option price $\rightarrow$ recovery probability (more info)

Recovery as a test of EMH under the joint-hypothesis problem

Traditional difficulty

Any empirical test of market efficiency is a joint test of efficiency and an assumed asset-pricing model. Rejection does not tell us whether markets are inefficient or the pricing model is misspecified.

Ross-style contribution

By recovering the pricing kernel directly from option prices, one can derive restrictions on admissible return predictability without first positing a parametric consumption model or historical factor structure.

- Recovery is therefore presented as a more model-light route to testing whether observed Sharpe ratios or forecasting performance are too large to be reconciled with efficient pricing.

Hansen-Jagannathan bound from the recovered kernel

Bound

Once the pricing kernel ϕ is recovered, its volatility bounds any Sharpe ratio consistent with efficient pricing:

$$\sigma(\phi) \geq e^{-rT} \frac{\mu}{\sigma}.$$

The original exercise obtains a recovered-kernel standard deviation near 0.3264, interpreted as an upper bound on the Sharpe ratio of any admissible strategy in that environment.

The Hansen-Jagannathan (H-J) Bound

From Recovery to Risk Limits

- In the previous section, we recovered the pricing kernel ϕ from option price data.
- **Goal**: Calculate the volatility of ϕ to set a limit on Sharp ratio.

The H-J Bound Condition

$$\sigma(\phi) \geq (e^{-rT}) \frac{\mu}{\sigma}$$

- $\sigma(\phi)$: The volatility of the pricing kernel (market's sensitivity to risk).
- $\frac{\mu}{\sigma}$: The Sharpe Ratio (Excess return μ divided by risk σ).

Empirical Results (From Table 3)

- Variance of the recovered kernel: $\sigma^2(\phi) = 0.1065$
- Annualized Standard Deviation: $\sigma(\phi) = 0.3264$
- $\Rightarrow$ **0.3264 is the upper limit for the Sharpe Ratio** of any investment strategy in an efficient market.

Bound on predictive R^2

Suppose excess returns satisfy

$$x_t = \mu(I_t) + \varepsilon_t,$$

where $\mu(I_t)$ is the conditional expected return and ε_t is orthogonal noise.

Variance decomposition and bound

$$\sigma^2(x_t) = \sigma^2\big(\mu(I_t)\big) + \sigma^2(\varepsilon_t) \le E[\mu^2(I_t)] + \sigma^2(\varepsilon_t),$$

so that

$$R^2 = \frac{\sigma^2(\mu(I_t))}{\sigma^2(x_t)} \le \frac{E[\mu^2(I_t)]}{\sigma^2(x_t)} \le e^{2rT}\sigma^2(\phi).$$

- In the original lecture notes, this implies a rough upper bound near 10% for return predictability once the recovered kernel variance is around 0.1.

Takeaway of the Ross block

What Ross gives us

- A beautiful decomposition of prices into discounting, state dynamics, and risk adjustment.
- A Perron–Frobenius route from Arrow prices to a candidate natural law.
- A bridge from option data to restrictions on Sharpe ratios and predictability.

What Ross does *not* give us

- A fully robust empirical recovery procedure in realistic incomplete and noisy markets.
- A general solution once long-run martingale components contaminate the link between Q and P.

Hence Ross is best understood as the conceptual doorway to the more operator-theoretic framework developed in the rest of Part II.

Roadmap of this supplement

1. Generalized Recovery as an early broadening of the recovery question
2. Formula bridge: from Ross ratio form to long-term factorization
3. BHS as the first systematic critique of exact recovery
4. Garman and Linetsky: semigroup kernels, Green's functions, and path integrals
5. Hansen–Scheinkman: multiplicative functionals, semigroup pricing, and the eigenproblem
6. Qin–Linetsky: Borel right processes, Feynman–Kac operators, and short-rate models
7. Dillschneider–Maurer: continuous-state functional recovery
8. Final return to the pricing operator M

Jensen–Lando–Pedersen (2019): Generalized Recovery

Let Q_t denote the Arrow-state price operator at horizon t:

$$Q_t f(x) = \mathbb{E}_x[S_t f(X_t)]\,.$$

Ross-style recovery concentrates on a one-step homogeneous relation. Generalized Recovery starts from the *family* $\{Q_t\}_{t\in\mathcal{T}}$ and asks whether cross-maturity consistency can add identification power:

$$\{Q_t\}_{t\in\mathcal{T}} \quad \Longrightarrow \quad \text{restrictions on dynamics and valuation from multiple maturities.}$$

A useful schematic contrast is:

Ross: $Q_1 \rightsquigarrow (P, \phi)$ vs. Generalized Recovery: $\{Q_t\}_{t\in\mathcal{T}} \rightsquigarrow$ broader Markov identification problem.

- identification can come from richer state-price information,
- not only from stricter transition structure,
- this already loosens the narrow one-step homogeneous Markov perspective.

Bridge in formulas: from Ross to long-term factorization

Ross imposes a transition-independent pricing kernel:

$$q_{ij} = \phi_{ij} f_{ij}, \qquad \phi_{ij} = \delta \frac{h_j}{h_i}, \qquad q_{ij} = \delta \frac{h_j}{h_i} f_{ij}.$$

A continuous-time operator version of the same ratio logic is the long-term factorization preview:

$$S_t = e^{-\lambda t} \frac{\pi(X_0)}{\pi(X_t)} H_t.$$

Thus the formal bridge is:

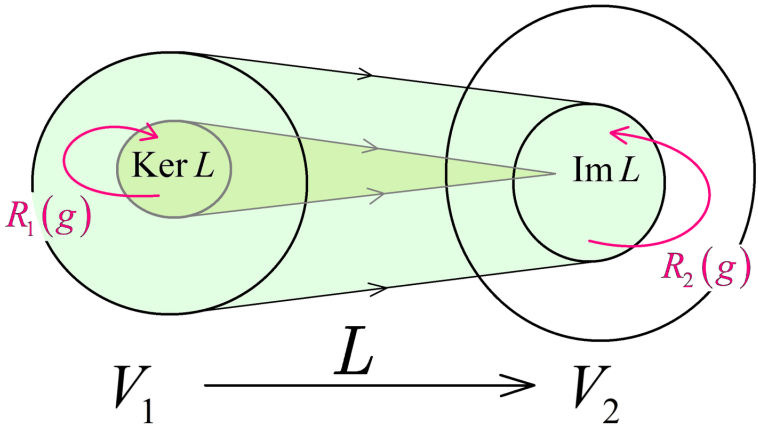
$$\underbrace{\delta\, h_j/h_i}_{\text{Ross ratio term}} \quad \Longleftrightarrow \quad \underbrace{e^{-\lambda t}\, \pi(X_0)/\pi(X_t)}_{\text{HS ratio term}} \times \underbrace{H_t}_{\text{martingale distortion}} .$$

Exact recovery is possible only when the martingale distortion is absent or trivial.

A useful algebraic analogy: the first isomorphism theorem

For a linear map $L : V_1 \to V_2$, the fundamental homomorphism theorem says

$$V_1/\ker L \cong \operatorname{Im} L, \qquad [v] \mapsto L(v).$$



Borovička–Hansen–Scheinkman (2016): misspecified recovery

Once the general factorization

$$S_t = e^{-\lambda t}\frac{\pi(X_0)}{\pi(X_t)}H_t$$

is taken as the benchmark, Perron–Frobenius recovery identifies the transformed pricing semigroup

$$\widetilde{M}_t f(x) = e^{\lambda t}\frac{1}{\pi(x)}M_t(\pi f)(x),$$

not automatically the physical semigroup. If H_t is nontrivial, then

$$\widetilde{M}_t \neq P_t$$

in general, where P_t is the physical transition semigroup. Hence the recovered measure is typically a *long-run valuation measure*:

$$\text{PF-recovered dynamics} \neq \text{subjective / physical dynamics.}$$

This is the conceptual break: exact recovery is not generic; prices already embed an irreducible distortion.

Why the critique forces a more general framework

BHS turns the original recovery question into a deeper structural question:

What is the object actually recovered by spectral pricing methods?

To answer this, we need a framework in which:

- discounting is dynamic and horizon-consistent,
- change of measure is represented internally,
- long-horizon asymptotics are governed by an eigenstructure,
- and the permanent distortion can be isolated mathematically.

This is exactly what the semigroup approach supplies:

$$M_t f(x) = \mathbb{E}_x[S_t f(X_t)], \qquad M_{t+u} = M_t M_u.$$

Garman (1985): toward a semigroup pricing theory

Dynamic pricing across horizons naturally suggests

$$\mathcal{P}_{t+s} = \mathcal{P}_t \mathcal{P}_s, \qquad \mathcal{P}_0 = I.$$

This is not merely formal. It means:

- pricing is linear,
- pricing is recursively composable,
- the full horizon structure is governed by one operator family rather than by isolated maturities.

In Markovian settings this becomes

$$\mathcal{P}_t f(x) = \mathbb{E}_x[S_t f(X_t)]\,,$$

so finite-state pricing matrices are only the lowest-dimensional case of a more general operator semigroup.

Linetsky (1998): abstract picture, work, and claim

$$\text{Risk-neutral pricing} \iff \text{Feynman–Kac representation} \iff \text{Path integral / Green's kernel}$$

- **Object**: contingent-claim values in a risk-neutral diffusion world, with special emphasis on path-dependent and multi-asset derivatives.
- **Representation**: option value as an integral over paths; equivalently, as an integral against a Green's function / transition kernel.
- **Claim**: path integrals and Green's functions are both a natural theoretical language and a practical computational tool in finance.
- **Contribution**: a unified bridge between martingale pricing, diffusion PDEs, semigroup kernels, and concrete valuation procedures.

$$\text{pricing over time} \Longrightarrow \text{kernel propagation} \Longrightarrow \text{semigroup}$$

Quick review: pricing as integration over paths

For a path-independent claim with payoff $F(S_T)$,

$$V(S,t) = e^{-r\tau}\,\mathbb{E}^{Q}_{t,S}\big[F(S_T)\big], \qquad \tau = T - t.$$

Setting $x = \log S$ and $dx = \mu\,dt + \sigma\,dW_t$, Linetsky rewrites the value as

$$V(S,t) = e^{-r\tau} \int_{x(t)=x}^{x(T)=x_T} F(e^{x_T})\, e^{-A_{BS}[x(\cdot)]}\, Dx(\cdot)\, dx_T,$$

with action functional

$$A_{BS}[x(\cdot)] = \int_t^T L_{BS}\, dt', \qquad L_{BS} = \frac{1}{2\sigma^2}(\dot{x} - \mu)^2.$$

- Every path carries a weight determined by the action.
- The path integral is the limit of finite-dimensional multiple integrals.
- In finance, this is simply the risk-neutral expectation written on path space.

Green's function as pricing kernel

The core kernel is the transition density / Green's function

$$K(x_T, T \mid x, t) = \int_{x(t)=x}^{x(T)=x_T} e^{-A_0[x(\cdot)]}\, Dx(\cdot).$$

Then pricing becomes an integral operator:

$$(\mathcal{P}_{t,T} f)(x) = e^{-r\tau} \int K_\mu(x_T, T \mid x, t)\, f(x_T)\, dx_T.$$

The kernel satisfies the Chapman–Kolmogorov semigroup property

$$K(x_3, t_3 \mid x_1, t_1) = \int K(x_3, t_3 \mid x_2, t_2) K(x_2, t_2 \mid x_1, t_1)\, dx_2,$$

so that

$$\mathcal{P}_{t,u} = \mathcal{P}_{t,s} \mathcal{P}_{s,u}.$$

Interpretation

The pricing operator is not an abstract symbol only: it is concretely a kernel integral operator, and the semigroup property is exactly kernel composition across horizons.

Different claims, different kernels

Vanilla / terminal-state claims

$$V(S,t) = e^{-r\tau} \int F(e^{x_T})\, K_\mu(x_T, T \mid x, t)\, dx_T.$$

Path-dependent claims with an additive functional If

$$I[x(\cdot)] = \int_t^T V(x(t'), t')\, dt',$$

then the relevant kernel becomes a Green's function with potential (or killing)

$$K_V(x_T, T \mid x, t) = \int \exp\left\{ -\int_t^T (L_0 + V)\, dt' \right\} Dx(\cdot),$$

with PDE

$$\frac{\sigma^2}{2} \partial_{xx} K_V - V(x,t) K_V = -\partial_t K_V.$$

Joint-state pricing via transforms For payoffs $F(x_T, I)$, Fourier or Laplace transforms produce a joint density

$$V(S,t) = e^{-r\tau} \iint F(e^{x_T}, \lambda)\, P(x_T, \lambda; T \mid x, t)\, d\lambda\, dx_T.$$

From one asset to many assets

For a D-dimensional risk-neutral diffusion,

$$dx^\mu = a^\mu dt + \sum_{a=1}^{D} \sigma_a^\mu \, dW_t^a,$$

Linetsky writes the action using the Lagrangian

$$L = \frac{1}{2} \sum_{\mu,\nu} g_{\mu\nu}(x,t)(\dot{x}^\mu - a^\mu)(\dot{x}^\nu - a^\nu) + r(x,t), \qquad (g_{\mu\nu}) = (g^{\mu\nu})^{-1}.$$

The corresponding generator is

$$H = \frac{1}{2} \sum_{\mu,\nu} g^{\mu\nu}(x,t)\partial_{\mu\nu} + \sum_{\mu} a^\mu(x,t)\partial_\mu - r(x,t),$$

and pricing still has the kernel form

$$V(x,t) = \int F(x_T)\, K(x_T, T \mid x, t)\, dx_T.$$

- Weighted Asian options: transform path averages into a joint density problem.
- Floating barrier and ladder-style barriers: encode state constraints through the relevant kernel.

Evaluation toolbox for path-integral pricing

How to evaluate the path integral

- **Exact analytical**: Gaussian path integrals, Van Vleck formula, changes of variables, time reparametrization, and projections.
- **Analytical approximation**: semiclassical expansion and WKB approximation.
- **Numerical**: Monte Carlo, quasi Monte Carlo, deterministic discretization, and finite-difference schemes viewed as path-integral discretizations.

The point is methodological: once pricing is written as a kernel over paths, different tools correspond to different ways of evaluating the same object.

Why this language is especially natural for options

Derivative pricing fits the path-space setup

$$V_t = e^{-r\tau}\mathbb{E}_t^Q[\text{known payoff functional}],$$

so the core task is already risk-neutral integration of a contractually specified payoff.

- Many derivative payoffs are genuinely path-dependent, so the path formulation is structural rather than decorative.
- For general assets such as equity, a kernel representation alone is not enough: dividend growth, discounting, beliefs, and equilibrium structure still have to be specified.
- This is why the path-integral toolkit feels especially natural for options and exotic derivatives.

Hansen–Scheinkman (2009): multiplicative functionals

Let X be a Markov state process on a general state space. The stochastic discount factor S is modeled as a positive **multiplicative functional**:

$$S_{t+u} = S_t\big(S_u \circ \theta_t\big), \qquad S_0 = 1.$$

A semimartingale representation makes the object fully dynamic:

$$d\log S_t = \beta(X_t)\,dt + \alpha(X_t)\,dW_t + \int \gamma(X_{t-}, y)\,\widetilde{N}(dt, dy).$$

Hence the same object can simultaneously encode:

- discounting,
- risk adjustment,
- cash-flow growth,
- and change of measure.

Why multiplicative functionals are theoretically central

The multiplicative property gives intertemporal consistency at the process level:

$$S_{t+u} f(X_{t+u}) = S_t \Big[(S_u f(X_u)) \circ \theta_t \Big].$$

Therefore the induced pricing operators

$$M_t f(x) = \mathbb{E}_x [S_t f(X_t)]$$

form a semigroup:

$$M_{t+u} = M_t M_u.$$

This is the structural reason spectral methods become available:

- positive eigenvalues govern long-run discounting,
- positive eigenfunctions capture state dependence,
- martingale components capture permanent measure distortion.

Hansen–Scheinkman (2009): eigenproblem and generator

Given the pricing semigroup $\{M_t\}_{t\geq 0}$, HS study the positive eigenfunction problem

$$M_t\pi(x) = e^{-\lambda t}\pi(x), \qquad t \geq 0.$$

At the generator level this becomes

$$\mathcal{G}\pi = -\lambda\pi,$$

where $\mathcal{G}$ is the extended generator of the pricing semigroup. Interpretation:

$\lambda =$ asymptotic discount rate / long bond yield, $\qquad \pi =$ positive state-dependent eigenfunction.

The dominant eigenpair governs long-horizon valuation asymptotics.

Hansen–Scheinkman (2009): long-term factorization

The stochastic discount factor admits the long-term factorization

$$S_t = e^{-\lambda t} \frac{\pi(X_0)}{\pi(X_t)} H_t,$$

where H_t is a positive martingale under appropriate conditions. This separates three distinct objects:

$$\underbrace{e^{-\lambda t}}_{\text{long-run discounting}} \times \underbrace{\frac{\pi(X_0)}{\pi(X_t)}}_{\text{transitory state term}} \times \underbrace{H_t}_{\text{permanent martingale distortion}}.$$

Thus long-term pricing is not just a ratio representation; it is a ratio representation *plus* a permanent martingale channel.

Qin–Linetsky (2014/2016, Operations Research): the state process

Qin and Linetsky recast the literature in a broader probabilistic language. Instead of working only with convenient Markov diffusions, they place the state process in the class of **Borel right processes**.

X is a Borel right process $\implies$ the semigroup / generator machinery survives on a very general state s

This matters because it reformulates earlier work in a unified way:

- Ross recovery,
- HS factorization,
- long-term pricing asymptotics,

all become statements about positive eigenfunctions of Markovian pricing operators on general state spaces.

Qin–Linetsky (2014/2016, Operations Research): Feynman–Kac operators

The key operator becomes the **Feynman–Kac pricing semigroup**:

$$\mathcal{P}_t^r f(x) = \mathbb{E}_x \left[e^{-\int_0^t r(X_s)\, ds} f(X_t) \right].$$

This identifies short-rate pricing as a special operator case. At the generator level,

$$\mathcal{G}^r = \mathcal{A} - r,$$

where $\mathcal{A}$ is the generator of the state process and r is the short-rate function. Hence traditional short-rate models are naturally rewritten inside the same spectral framework:

$$\mathcal{P}_t^r \pi = e^{-\lambda t} \pi.$$

This is why the paper is more than an extension: it reorganizes pricing theory around Feynman–Kac generators and positive eigenfunctions.

Qin–Linetsky: why this paper is the clean bridge

The Operations Research paper contributes four conceptual upgrades:

1. from Markov diffusion examples to general Borel right processes,
2. from one-off pricing formulas to Feynman–Kac operator semigroups,
3. from separate results to one eigenfunction-based reformulation,
4. from abstract recovery theory back to traditional short-rate models and long bond pricing.

Schematic bridge:

Ross recovery $\subset$ HS long-term factorization $\subset$ positive eigenfunctions of general Markovian pricing operato

This is why the Qin–Linetsky paper is the sharpest theoretical bridge in the literature.

Dillschneider–Maurer (2019, JEDC): functional Ross recovery

Functional Ross Recovery replaces Perron–Frobenius matrix theory with Perron–Frobenius operator theory on continuous state spaces:

$$(\mathcal{P}\varphi)(x) = \rho\,\varphi(x).$$

Main conclusion:

- Ross-style recovery can be extended from discrete states to continuous states,
- the recovery object is again pinned down by a positive eigenfunction problem,
- but identification remains sensitive to state-space specification and operator restrictions.

So the paper preserves the structural elegance of Ross while making explicit that recovery becomes an operator problem, not merely a matrix problem.

Return to the pricing operator M

The key object of the whole block is still

$$M_t f(x) = \mathbb{E}_x[S_t f(X_t)] \,.$$

It should be read simultaneously as:

1. an **algebraic object**: linear, composable, spectrally analyzable;
2. a **probabilistic object**: it changes measure and distorts dynamics;
3. an **identification bottleneck**: prices are only a non-faithful representation of the underlying transition matrix.

Thus the deep issue is not whether one can write down an eigenproblem, but whether the recovered eigenstructure can be interpreted as physical dynamics.

Takeaway of the whole block

Takeaway

- The current consensus is not that exact recovery succeeds, but that *exact recovery is generally impossible* outside very strong restrictions.
- The recovered object is inevitably distorted by a permanent martingale component or by missing structural information.
- Prices are therefore a *non-faithful representation* of the underlying probability transition matrix.
- The more general and more stable object is the pricing semigroup generated by a positive multiplicative functional.
- Garman, Linetsky, HS, Qin–Linetsky, and Dillschneider–Maurer together shift the focus from exact recovery to operator structure, kernels, long-term factorization, and the limits of identification.

The Party is Over? Long Live the King!

Part III-A

Ghosh, Julliard, and Taylor (2025, JFEC)

Six Questions as Hooks: What Does the Paper Actually Do?

- **How is the I-SDF recovered?** By solving a minimum relative entropy problem under no-arbitrage pricing restrictions.
- **Why KL divergence?** Because the objective penalizes the full distortion from the physical measure, not just variance.
- **What is inherited from GJT (2017)?** The same minimum-entropy recovery logic; this paper moves it from in-sample identification to rolling out-of-sample pricing.
- **How is the recovery linked to empirics?** The recovered object becomes a single-factor candidate SDF in standard cross-sectional tests.
- **What kind of object is recovered?** A strictly positive, nonlinear, strongly non-Gaussian I-SDF with large tail-risk content.
- **What does the paper find?** The abstract result is clear: the I-SDF delivers smaller out-of-sample pricing errors, better cross-sectional fit than leading multifactor models, and captures risk not spanned by standard factors.

Abstract-level empirical claims emphasized by the paper

- Very large annual alphas relative to standard factor models: 20–37%.
- A large Sharpe ratio for the information anomaly: about 1.1.
- The recovered I-SDF is highly positively skewed and leptokurtic.
- Roughly one third of observed risk premia are compensation for about 2.5% tail events.

Core Principle: Minimum Relative Entropy

Start from the excess-return Euler equation under the physical measure P:

$$0 = \mathbb{E}_P[M_t R_t^e].$$

Let

$$\frac{M_t}{\bar{M}} = \frac{dQ}{dP}, \qquad \bar{M} \equiv \mathbb{E}[M_t].$$

Then the pricing restrictions can be rewritten as

$$0 = \int \frac{M_t}{\bar{M}} R_t^e \, dP = \int R_t^e \, dQ = \mathbb{E}_Q[R_t^e].$$

The recovery problem is therefore:

$$\arg\min_Q D(Q\|P) \quad \text{s.t.} \quad \int R_t^e \, dQ = 0.$$

- The paper does not specify preferences or the identity of priced factors ex ante.
- It searches over all probability distortions that satisfy no-arbitrage pricing.
- The guiding principle is: choose the pricing measure that adds the least extra information relative to history.

Sample Form and Rolling Out-of-Sample Design

Sample analogue of the minimum-entropy problem (Eq. 3)

$$\arg\min_{\{M_t\}_{t=1}^T} \frac{1}{T}\sum_{t=1}^{T} M_t \ln M_t \quad \text{s.t.} \quad \frac{1}{T}\sum_{t=1}^{T} M_t R_t^e = 0.$$

Fenchel-dual solution (Eqs. 4–5)

$$\hat{M}_t = \frac{T e^{\hat{\theta}' R_t^e}}{\sum_{s=1}^{T} e^{\hat{\theta}' R_s^e}}, \qquad \hat{\theta} = \arg\min_{\theta} \frac{1}{T}\sum_{t=1}^{T} e^{\theta' R_t^e}.$$

The Fenchel-dual solution also has the form of a partition function in statistical physics.

- **Sample form**: the constraints are the sample Euler equations; the recovered object is a nonlinear function of returns.
- **Rolling out-of-sample**: estimate $\hat{\theta}_{T_i}$ in each training window, then construct $M(\hat{\theta}_{T_i}, R_t^e)$ over the next s periods.
- The baseline design uses a 30-year rolling window and annual rebalancing ($s = 12$ months).
- The exponential form guarantees strict positivity both in sample and out of sample.

KL Divergence: Definition and Why It Is Used

Definition

$$D(A\|B) = \int \ln\left(\frac{dA}{dB}\right) dA, \qquad D(Q\|P) = \int \frac{dQ}{dP} \ln\left(\frac{dQ}{dP}\right) dP.$$

- $D(A\|B) \geq 0$, with equality only when $A = B$.
- It is not symmetric: generally $D(A\|B) \neq D(B\|A)$.
- In this paper it measures the information cost of distorting the physical measure P into a pricing measure Q.
- Relative to a quadratic divergence, KL is not just a variance criterion: it penalizes the entire probability distortion.
- The log form naturally supports a strictly positive recovered SDF.
- The paper also stresses that entropy-based reweighting is especially useful when tail events matter.

GJT (2017) already used this minimum-entropy principle to recover SDF objects and derive tighter entropy bounds. The JFEC paper keeps the same principle and asks a different question: does the recovered SDF work out of sample as a cross-sectional pricing benchmark?

I-SDF versus Tangency Portfolio

The paper distinguishes the minimum-entropy SDF from a minimum-variance tangency portfolio.

Cumulant expansion

$$D(P\|\hat{Q}) = \frac{\kappa_2^{\hat{M}}}{2!} + \frac{\kappa_3^{\hat{M}}}{3!} + \frac{\kappa_4^{\hat{M}}}{4!} + \cdots$$

so entropy minimization penalizes a weighted sum of all cumulants, not just variance.

- If the true pricing kernel is lognormal, higher-order cumulants vanish, so minimum relative entropy collapses to a variance criterion.
- In that special case, the I-SDF degenerates to the tangency-portfolio logic.
- Outside that case, the I-SDF and tangency portfolio are conceptually different objects.

In-sample linear projection of the I-SDF

$$\omega_{TP} = \frac{[\mathrm{Var}(R)]^{-1}\mathbb{E}(R^e)}{\iota'[\mathrm{Var}(R)]^{-1}\mathbb{E}(R^e)}, \qquad b = \frac{[\mathrm{Var}(R)]^{-1}\mathrm{Cov}(R,M)}{\iota'[\mathrm{Var}(R)]^{-1}\mathrm{Cov}(R,M)}.$$

Using the Euler equation $\mathbb{E}(R^e) = -\mathrm{Cov}(R,M)/\mathbb{E}(M)$, the paper shows that $b \equiv \omega_{TP}$ in sample. Hence the tangency portfolio is the linear projection of the I-SDF, not the I-SDF itself.

Sampling Interpretation: Multinomial Weights and Nonparametric MLE

The paper gives the recovery a sampling interpretation. Let $\{\tilde{q}_t\}_{t=1}^T$ be multinomial probabilities assigned to the historical sample points.

Likelihood of the sampled weights

$$L(\{\tilde{q}_t\}) = \frac{N!}{(N\tilde{q}_1)! \cdots (N\tilde{q}_T)!}.$$

Using Stirling's formula, as $N \to \infty$,

$$\log L(\{\tilde{q}_t\}) \to -\sum_{t=1}^{T} \tilde{q}_t \log \tilde{q}_t.$$

- Therefore, under the same pricing restrictions, maximizing the multinomial likelihood is equivalent to minimizing relative entropy.
- This is the paper's nonparametric maximum-likelihood interpretation of entropy minimization.
- The key point is separate from the constraints: the constraints in Eq. (3) come from the sample Euler equations, while Stirling's formula is used to justify the sampling / likelihood interpretation.

Empirical: Data and Rolling Design

- **Baseline frequency and sample**: monthly returns, main evaluation period 1963:07–2017:06. With a 30-year training window, data are required from 1933:07.
- **Base assets, low-dimensional design**: 15 portfolios built from 5 industries and the smallest/largest deciles of size, book-to-market, momentum, short-term reversal, and long-term reversal.
- **Base assets, high-dimensional design**: 135 portfolios spanning size, BM, momentum, STR, LTR, profitability, investment, accruals, new issues, beta, variance, residual variance, and earnings-to-price.
- **Data source**: Kenneth French Data Library for U.S. equity portfolios and the 1-month T-bill rate as the risk-free rate proxy.
- **Rolling construction**: estimate $\hat{\theta}$ in each training window and use it to generate the I-SDF over the next 12 months.
- **High-dimensional extension**: the paper adopts the penalized form

$$\hat{\theta} = \arg\min_{\theta} \frac{1}{T} \sum_{t=1}^{T} e^{\theta' R_t^e} + \gamma_N \|\theta\|_1,$$

following Qiu and Otsu (2022).

Empirical Testing Framework

The recovered I-SDF is evaluated exactly as a standard candidate asset-pricing factor.

Two-pass regressions

$$R_t^e = a + BF_t + \varepsilon_t,$$
$$\mu = z\iota + B\gamma + \alpha.$$

- **Competing models**: CAPM, FF3, FF5, HXZ, KNS PCA-SDF, HJ-SDF, and the recovered I-SDF.
- **Estimated objects**: betas, factor prices, and pricing errors, with Shanken-corrected standard errors.
- **Reported fit statistics**: adjusted OLS R^2, adjusted GLS R^2, Shanken's T^2, and

$$q = \alpha'(ySy)^+\alpha.$$

- **Reading the statistics**: GLS R^2 and q are especially informative because they directly locate the model relative to the mean–variance frontier.
- The empirical link is therefore simple: recover a time series for the I-SDF, then test it as a single-factor SDF in the standard cross-sectional framework.

I-SDF Properties: Time-Series Pattern, Distribution, and Tail Risk

Time-series and distributional shape

- The I-SDF is low in expansions and high in recessions, especially when recessions coincide with major stock-market declines.
- Table 6 reports a sharply non-Gaussian object:

median	0.85
volatility	0.80
skewness	4.52
excess kurtosis	35.5

- The empirical object is therefore not a mild perturbation of a linear factor portfolio; it is a strictly positive, tail-sensitive pricing kernel.

Tail-risk decomposition

- Top 5% of I-SDF realizations explain 33.0% of the market risk premium.
- Bottom 5% of market returns explain 50.4% of the market risk premium.
- The joint tail of the I-SDF and market returns explains 42.7% of the market premium using only about 1.2% of states.
- For the 55 equity test assets, the joint tail explains roughly 37.4%–60.7% of average premia.

Interpretation: a large fraction of priced risk sits in rare states, so the recovered I-SDF is best read as a nonlinear tail object rather than as a Gaussian benchmark factor.

Cross-sectional Empirical Results: How to Read Table 1

Table 1 asks: if one recovers an I-SDF separately for each test cross section, how well does it price that same cross section out of sample?

Test assets	**OLS** R^2	**GLS** R^2	T^2 **p-value**	q
25 FF portfolios	84.5%	72.6%	0.384	0.045
10 momentum portfolios	94.1%	81.6%	0.467	0.013
55 anomaly/industry assets	91.4%	89.9%	0.907	0.013

- Relative to CAPM, FF3, FF5, HXZ, and KNS, the I-SDF produces higher OLS and GLS fit, smaller pricing errors, and a smaller distance to the mean–variance frontier.
- HJ-SDF is the closest comparator in point estimates, but it generates negative SDF realizations, unlike the I-SDF.
- This table is the paper's strongest evidence that the entropy-recovered SDF can behave like a serious out-of-sample pricing benchmark.

Cross-sectional Empirical Results (Continued)

Table 3: One unique I-SDF from 15 base portfolios

- The paper then stops re-estimating a separate I-SDF for each test set.
- Instead, it recovers one unique I-SDF from 15 base portfolios and uses it to price broader test cross sections.
- This turns the exercise into a test of transferability and robustness.
- On 55 test assets, the unique I-SDF reaches 74.7% OLS R^2 and 43.1% GLS R^2, well above FF3, FF5, and HXZ in the same exercise.

Table 4: Compressing FF3/FF5/HXZ into single-SDF form

- Once FF3, FF5, and HXZ are also forced into a single-SDF representation, their cross-sectional performance weakens sharply.
- This is the paper's way of comparing single-factor SDF objects to single-factor SDF objects, rather than comparing one SDF to unrestricted multifactor regressions.

Table 5 and Table 8

- In the 135-portfolio high-dimensional design, the penalized I-SDF remains workable, especially relative to other one-factor implied SDFs.
- Standard factors explain only a small fraction of I-SDF variation: FF3, FF5, and HXZ explain about 4.4%, 10.0%, and 12.6%, respectively.
- This is the empirical basis for the paper's information anomaly: the recovered I-SDF contains pricing information not spanned by standard factors.

Review: What the Paper Establishes

- The paper's organizing questions all point back to one principle: recover the pricing kernel by minimum relative entropy under no-arbitrage restrictions.
- KL divergence is not auxiliary decoration here; it is the criterion that determines which pricing distortion is admissible and which one is minimal.
- The recovered object is a strictly positive, nonlinear, out-of-sample I-SDF, not merely a linear factor portfolio.
- The sampling argument shows that entropy minimization also has a nonparametric likelihood interpretation on the historical support.
- Empirically, the I-SDF delivers strong cross-sectional performance relative to standard benchmarks and remains informative when the exercise is made harder: one recovered SDF, broader test sets, and even high-dimensional anomaly spaces.
- The paper's substantive contribution is therefore twofold: a principled recovery method and a strong empirical benchmark for the latent pricing kernel.

Part III-B

Ran and Martin: Moments of the SDF

Why introduce the joint CGF?

The paper studies the joint cumulant-generating function

$$\kappa_t(\theta_1, \theta_2) = \log E_t \left[(M_{t+1} R_{f,t+1})^{\theta_1} \left(\frac{R_{t+1}}{R_{f,t+1}} \right)^{\theta_2} \right].$$

Three restrictions follow directly from asset-pricing identities:

$$\kappa_t(0,0) = 0, \qquad \kappa_t(1,0) = 0, \qquad \kappa_t(1,1) = 0.$$

The motivation is that the joint CGF collects three objects in one place:

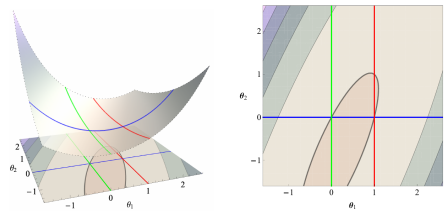
$$\kappa_t(0,\theta), \qquad \kappa_t(1,\theta), \qquad \kappa_t(\theta,0).$$

- $\kappa_t(0,\theta)$: physical-measure return CGF;
- $\kappa_t(1,\theta)$: risk-neutral return CGF;
- $\kappa_t(\theta,0)$: SDF CGF.

The key technical property is convexity of $\kappa_t(\theta_1, \theta_2)$ in (θ_1, θ_2).

The joint CGF surface and the three slices

$$\kappa_t(0,\theta) = \log E_t\left[\left(\frac{R_{t+1}}{R_{f,t+1}}\right)^{\theta}\right], \quad \kappa_t(1,\theta) = \log E_t^*\left[\left(\frac{R_{t+1}}{R_{f,t+1}}\right)^{\theta}\right],$$
$$\kappa_t(\theta,0) = \log E_t[(M_{t+1}R_{f,t+1})^{\theta}].$$



(a) The CGF surface (b) Contour plot

Figure 1: The cumulant-generating function (1). The thick contour in the right panel is the zero contour.

- The green slice $\kappa_t(0,\theta)$ is observed under the physical measure P.
- The red slice $\kappa_t(1,\theta)$ is observed under the risk-neutral measure Q.

Moment bounds from convexity of the joint CGF

For $\theta < 0$ or $\theta > 1$,

$$\kappa_t(\theta, 0) \geq \sup_{y \in \mathbb{R}} \left\{ \theta\, \kappa_t(1, y) + (1 - \theta)\, \kappa_t\left(0, \frac{\theta}{\theta - 1} y\right) \right\}.$$

For $\theta \in (0, 1)$,

$$\kappa_t(\theta, 0) \leq \inf_{y \in \mathbb{R}} \left\{ \theta\, \kappa_t(1, y) + (1 - \theta)\, \kappa_t\left(0, \frac{\theta}{\theta - 1} y\right) \right\}.$$

The three relevant points are collinear:

$$(\theta, 0), \qquad \left(0, \frac{\theta}{\theta - 1} y\right), \qquad (1, y).$$

Hence the bounds are obtained by evaluating the convex surface along the line joining an observable point on the P-slice and an observable point on the Q-slice.

Information entropy bounds: definition, computation, interpretation

Define two entropy metrics for any positive random variable X:

$$L_t^{(1)}(X) = E_t[X \log X] - (E_t X)\log(E_t X), \qquad L_t^{(2)}(X) = \log(E_t X) - E_t[\log X].$$

For $X = M_{t+1}R_{f,t+1}$ and $E_t(M_{t+1}R_{f,t+1}) = 1$,

$$L_t^{(1)} = E_t[(MR_f)\log(MR_f)], \qquad L_t^{(2)} = -E_t \log(MR_f).$$

Result 4 gives

$$L_t^{(1)}(MR_f) \geq \sup_{y\in\mathbb{R}} \left\{ yE_t^* \log \frac{R_{t+1}}{R_{f,t+1}} - \log E_t \left[\left(\frac{R_{t+1}}{R_{f,t+1}} \right)^y \right] \right\},$$

$$L_t^{(2)}(MR_f) \geq \sup_{y\in\mathbb{R}} \left\{ yE_t \log \frac{R_{t+1}}{R_{f,t+1}} - \log E_t^* \left[\left(\frac{R_{t+1}}{R_{f,t+1}} \right)^y \right] \right\}.$$

These are endpoint slopes of the SDF CGF:

$$L_t^{(2)}(MR_f) = -\left.\frac{\partial \kappa_t(\theta,0)}{\partial\theta}\right|_{\theta=0}, \qquad L_t^{(1)}(MR_f) = \left.\frac{\partial \kappa_t(\theta,0)}{\partial\theta}\right|_{\theta=1}.$$

Higher-order cumulants and non-Gaussianity

Let κ_n denote the n-th cumulant of $\log(M_{t+1}R_{f,t+1})$. Equation (42) gives

$$L^{(1)}(MR_f) - L^{(2)}(MR_f) = \sum_{n=3}^{\infty} \frac{n-2}{n!}\,\kappa_n.$$

Expanding the first few terms,

$$L^{(1)} - L^{(2)} = \frac{1}{6}\kappa_3 + \frac{1}{12}\kappa_4 + \frac{1}{40}\kappa_5 + \frac{1}{180}\kappa_6 + \frac{1}{1008}\kappa_7 + \cdots$$

- If $\log(MR_f)$ is Gaussian, then $\kappa_n = 0$ for all $n \geq 3$, so $L^{(1)}(MR_f) = L^{(2)}(MR_f)$.
- The gap between the two entropy bounds is entirely driven by higher-order cumulants.
- The first entropy metric is more sensitive to the right tail of the SDF distribution.

Conditional vs. unconditional implementation

The paper uses an *unconditional* implementation, but the option-implied inputs are built from *conditional* risk-neutral CGFs.

Physical-measure CGF:

$$\hat{\kappa}(0,\theta) = \log \frac{1}{T} \sum_{t=0}^{T-1} \left(\frac{R_{t+1}}{R_{f,t+1}} \right)^{\theta}.$$

Conditional risk-neutral CGF from option prices:

$$\kappa_t(1,\theta) = \log \left\{ 1 + \theta(\theta-1) \left[\int_0^1 K^{\theta-2} \, \mathrm{put}_t(KR_{f,t+1}) \, dK + \int_1^{\infty} K^{\theta-2} \, \mathrm{call}_t(KR_{f,t+1}) \, dK \right] \right\}.$$

Unconditional risk-neutral CGF:

$$\hat{\kappa}(1,\theta) = \log \frac{1}{T} \sum_{t=1}^{T} \exp\{\kappa_t(1,\theta)\}.$$

Data and measurement:

- monthly US stock market returns, 1872–2022 and 1946–2022;
- month-end SPX option prices, 1996–2022; one-month horizon in the baseline;
- most sampling uncertainty comes from realized returns rather than the option-implied side.

Figure 4: singularity in SDF moment bounds

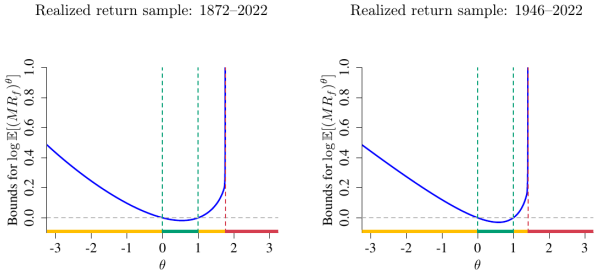


Figure 4: Convexity bounds for the moments of the SDF. The blue curves are lower bounds on SDF moments when $\theta < 0$ or $\theta > 1$ and upper bounds when $\theta \in (0, 1)$. Shading along the axis indicates the green, yellow and red regions as discussed in Section 3.2.

the singularity. As noted earlier, this phenomenon can emerge in a variety of equilibrium

Takeaways and comments

Takeaways

- The joint CGF provides a unified object linking the physical distribution, the risk-neutral distribution, and the SDF.
- The paper delivers sharp and well-behaved bounds for intermediate moments, especially for $\theta \in (0, 1)$.
- Entropy bounds are endpoint-slope objects and connect directly to higher-order cumulants and tail behavior.
- Empirically, the singularity appears before $\theta = 2$: about 1.72 in 1872–2022, 1.38 in 1946–2022, and 1.44 in 1996–2022.

Comments

- For a broader lecture on SDFs, recovery, and HS2009, the natural emphasis is on the CGF geometry and the entropy objects, not on the full empirical block.
- A brief empirical punchline is enough: singularity is near, while entropy-type objects remain well-behaved and informative.
- A useful discussion point is whether singularity should be interpreted as evidence against variance-based summaries, or more broadly as evidence of severe non-Gaussianity in the SDF.

Part III-C

Kelly, Malamud, and Pedersen (2023)

Prediction matrix: the primitive object

$$\Pi = \mathbb{E}(R_{t+1} S_t'), \qquad \Pi_{ij} = \mathbb{E}(R_{i,t+1} S_{j,t}).$$

- Index convention: i denotes the return asset (row index), while j denotes the signal asset (column index).
- Diagonal entries: own-signal predictability.
- Off-diagonal entries: cross-predictability.
- The standard simple factor uses only average own-predictability through the trace:

$$F^e_{t+1} = \sum_j S_{j,t} R_{j,t+1}, \qquad \mathbb{E}(F^e_{t+1}) = \mathrm{tr}(\Pi).$$

The usual own-predictor criterion discards both cross-predictability and heterogeneity on the diagonal.

Linear trading strategies and Proposition 1

$$w_t' = S_t' L, \qquad R_{t+1}^{w_t} = S_t' L R_{t+1}.$$

- The position matrix $L \in \mathbb{R}^{N \times N}$ maps all signals into all portfolio positions.
- The simple factor is the special case $L = I$.
- Proposition 1 links prediction to portfolio choice through a closed-form expected-return expression.

Proposition 1 (Return of Linear Strategies)

$$\mathbb{E}(R_{t+1}^{w_t}) = \mathbb{E}(S_t' L R_{t+1}) = \operatorname{tr}(L\Pi).$$

The trace term is the expected excess return. This is the exact quantity optimized later.

Portfolio construction is an optimization problem

Objective (equation 9)

$$\max_{L:\,\|L\|\le 1} \mathbb{E}(S_t' L R_{t+1}).$$

- The constraint is on the position matrix itself: portfolio construction is posed directly over L.
- Without a norm bound, the problem is ill-posed because scaling L scales expected return.
- When $\Sigma_{R,t} = \sigma^2 I$, the same problem is equivalent to a robust mean-variance problem.

Robust mean-variance form (equations 10–11)

$$\max_{L} \mathbb{E}(S_t' L R_{t+1}) \quad \text{s.t.} \quad \max_{S:\,\mathrm{Var}_t(S' R_{t+1})\le 1} \mathrm{Var}_t(S' L R_{t+1}) \le 1.$$

$$\max_{S:\,\mathrm{Var}_t(S' R_{t+1})\le 1} \mathrm{Var}_t(S' L R_{t+1}) = \|L\|^2 \quad \Longleftrightarrow \quad \|L\| \le 1.$$

Equivalently, the problem optimizes a linear operator L over the admissible unit ball in the underlying vector space.

Proposition 3: the optimizer becomes a singular-value problem

Closed-form optimizer

$$L^* = (\Pi'\Pi)^{-1/2}\Pi', \qquad \max_{L:\,\|L\|\le 1} \mathbb{E}(S_t' L R_{t+1}) = \sum_{i=1}^{N} \bar{\lambda}_i.$$

- Proposition 3 converts the optimization problem into spectral objects of Π.
- The maximal expected excess return is the sum of the singular values of Π.
- Only after this step does SVD become the natural implementation device.

Apply SVD

$$\Pi = U\bar{\Lambda}V', \qquad L^* = VU' = \sum_{k=1}^{N} v_k u_k'.$$

Optimization first, decomposition second.

Principal portfolios

$$L_k = v_k u_k', \qquad PP_{t+1}^k = S_t' v_k \, u_k' R_{t+1}.$$

- Each principal portfolio is a rank-one linear strategy.
- It trades the return portfolio u_k using the signal portfolio v_k.
- Proposition 4: the expected return of PP_k equals the k-th singular value.

$$\mathbb{E}(PP_{t+1}^k) = \bar{\lambda}_k, \qquad \mathbb{E}\left(\sum_{k=1}^{N} PP_{t+1}^k\right) = \sum_{k=1}^{N} \bar{\lambda}_k.$$

Principal portfolios are the building blocks of the optimal linear strategy.

Orthogonal projection: symmetric vs. antisymmetric prediction

$$\Pi = \Pi^s + \Pi^a, \qquad \Pi^s = \tfrac{1}{2}(\Pi + \Pi'), \qquad \Pi^a = \tfrac{1}{2}(\Pi - \Pi').$$

- Motivation: raw SVD on Π does not distinguish reciprocal predictability from directional predictability.
- Orthogonal projection separates the matrix into two subspaces with different economic content.
- The symmetric component organizes exposure-type predictability.
- The antisymmetric component organizes factor-neutral alpha.
- PEPs arise from eigenvectors of Π^s; PAPs arise from the antisymmetric component Π^a.

Projection converts a raw prediction matrix into a structured alpha/beta decomposition.

Dialogue between PCA and PPA (Table 1)

Table I

Analogy between PCA and PPA

This table shows five analogies between principal component analysis (PCA) and principal portfolio analysis (PPA) for the symmetric part of the prediction matrix. For PCA (PPA): (i) the variance (expected excess returns) of each component equals its eigenvalue; (ii) different components $k \neq l$ are orthogonal; (iii) the sum of the variances (returns) of individual securities equals that of the components, and also equals the trace of the variance-covariance matrix (prediction matrix); (iv) the top K components maximize variance (return) for orthonormal portfolios; and (v) component $k+1$ maximizes the variance (return) among all portfolios that are orthogonal to the first k portfolios.

	Principal Component Analysis	Principal Portfolio Analysis (Symmetric Part)
(i)	$\text{var}(R^{\pi_k}_{t+1}) \;=\; \lambda_k(\Sigma_R)$	$E(S_t^{w^s_k} R^{w^s_k}_{t+1}) \;=\; \lambda_k(\Pi^s)$
(ii)	$\pi_k' \Sigma_R \pi_l = 0$ i.e., $\text{cov}(R^{\pi_k}_{t+1}, R^{\pi_l}_{t+1}) \;=\; 0$	$(w^s_k)' \Pi^s w^s_l = 0$ i.e., $E(S_t^{w^s_k} R^{w^s_l}_{t+1}) \;+\; E(S_t^{w^s_l} R^{w^s_k}_{t+1}) = 0$
(iii)	$\sum_k \text{var}(R_{k,t+1}) = \sum_k \text{var}(R^{\pi_k}_{t+1}) = \text{tr}(\Sigma_R)$	$\sum_k E(S_{k,t} R_{k,t+1}) = \sum_k E(S_t^{w^s_k} R^{w^s_k}_{t+1}) = \text{tr}(\Pi^s)$
(iv)	$(\pi_k) = \arg\max_{\text{orthon.}\{x_k\}_{k=1}^K} \sum_k \text{var}(R^{x_k}_{t+1})$	$(w^s_k) = \arg\max_{\text{orthon.}\{x_k\}_{k=1}^K} \sum_k E(S_t^{x_k} R^{x_k}_{t+1})$
(v)	$\pi_{k+1} = \arg\max_{x \perp \{\pi_1, \ldots, \pi_k\}} \text{var}(R^x_{t+1})$	$w^s_{k+1} = \arg\max_{x \perp \{w^s_1, \ldots, w^s_k\}} E(S^x_t R^x_{t+1})$

State dynamics: static and dynamic opportunities

$$\Pi = \mathbb{E}(R_{t+1})\mathbb{E}(S_t') + \mathrm{Cov}(R_{t+1}, S_t').$$

Equation (38): static + dynamic decomposition.

- Static term: unconditional average-return information combined with average signals.
- Dynamic term: genuine timing information in the joint variation of signals and future returns.

$$\tilde{S}_{i,t} = S_{i,t} - \mathbb{E}(S_{i,t}), \qquad \mathbb{E}(R_{t+1}\tilde{S}_t') = \mathrm{Cov}(R_{t+1}, S_t').$$

Equation (39): demeaning isolates dynamic opportunities.

$$\mathbb{E}_t(R_{j,t+1}) = (1 + R_t^f)\,\mathrm{Cov}_t(R_{j,t+1}, -M_{t+1}) = \theta_t S_{j,t}.$$

Equation (40): pricing-kernel interpretation of the signal.

Asset-pricing implication: positivity of the prediction matrix

- If the signal is proportional to exposure to the pricing kernel, then expected returns satisfy

$$\mathbb{E}(R_{i,t+1} \mid \theta_t, S_t) = \theta_t S_{i,t}, \qquad \theta_t \geq 0.$$

- Under this condition, the prediction matrix must be symmetric and positive semidefinite.
- Equivalently: all PEPs must have non-negative expected returns and all PAPs must have zero expected returns.

Proposition 9

$$\Pi_{j,i} = \mathbb{E}(S_{i,t} R_{j,t+1}) = \mathbb{E}(\theta_t S_{i,t} S_{j,t}) = \Pi_{i,j}, \qquad w'\Pi w = \mathbb{E}(\theta_t [w' S_t]^2) \geq 0.$$

This yields an unconditional test of conditional asset-pricing restrictions.

Empirical implementation

Item	Empirical design
Setting	Base case: 25 Fama-French size-value portfolios, 20-day returns, and 20-day momentum signals.
Estimator	Rolling prediction matrix: $\hat{\Pi}_t = \frac{1}{120}\sum_{\tau=t-120}^{t-1} R_{\tau+1} S_\tau'$.
Training window	120 most recent non-overlapping return observations.
Outputs	PP from SVD of $\hat{\Pi}_t$; PEP/PAP from decompositions of $\hat{\Pi}_t^s$ and $\hat{\Pi}_t^a$.
Scope	Evidence spans U.S. equities, international equities, and global futures, with multiple lookback horizons.

The empirical object is always an estimated prediction matrix; the decomposition is recomputed in rolling out-of-sample fashion.

Empirical evidence: what the figures and tables show

Source in paper	Empirical content
Figure 1	Average singular values / eigenvalues align with realized out-of-sample returns of the corresponding PP, PEP, and PAP portfolios.
Figure 3A	PEP has a Sharpe ratio similar to the simple factor; PAP is higher; the combination of top PEPs and PAPs delivers more than twice the Sharpe ratio of the simple factor.
Figure 3B + Table 2	PEP has no significant alpha but strong loading on the simple factor and high R^2. PAP has small and insignificant factor loadings, low R^2, and significant alpha with t-statistic 4.42.
Interpretation	The evidence matches the theory: PEP behaves like priced exposure, while PAP behaves like factor-neutral alpha.

Prediction remains central empirically: the estimated matrix generates the ranking, the decomposition, and the interpretation.

Core contributions and comments

Core contributions

- Replaces asset-by-asset predictability with the matrix object $\Pi = \mathbb{E}(R_{t+1}S_t')$.
- Writes portfolio construction directly as an optimization problem over the linear operator L.
- Shows that the optimal solution is spectral: PP from SVD, PEP/PAP from orthogonal projection.
- Derives an unconditional asset-pricing restriction through symmetry and positive semidefiniteness of Π.

Comments

- The organizing principle is prediction, not factor construction.
- Optimization logically precedes decomposition; SVD is a consequence, not the starting point.
- The PCA analogy is precise only on the symmetric part of the prediction matrix.
- Empirically, the central bottleneck is estimating and regularizing $\hat{\Pi}_t$ in high dimension.

What's Our Overall Takeaway?

Overall Takeaways

1. **SDF Matters.** Most theoretical asset-pricing questions can be reframed as problems of *identification*, *approximation*, or *decomposition* of the stochastic discount factor. Under additional structure, the SDF can often be factorized, and in dynamic settings it induces pricing operators and *semigroups* linked to state dynamics.

2. **Portfolio Matters.** Portfolio choice is the tradable projection of priced information and solution for optimization. Could span to the "real" SDF.

3. **Entropy Matters: for nonlienar and tail-sensitive SDF.** While facing *the curse of dimension*, information entropy would offer new and (possibly) more efficient measurement rather than euclidean distance. Consider information geometry for SDF!

4. **Recovery Matters: identification how states transact.** Recovery shifts attention away from return forecasting alone. Starting from no-arbitrage prices, it asks what state prices reveal about beliefs, transition dynamics, and valuation.

5. **Spectral Matters: for long-term invariant dynamic structure.**

Open Questions and Research Directions

- **From representative agents to heterogeneous agents (HA) and their subjective beliefs.**
- **From average premia to tail states.**
- **From first-best to other best under economic constrains.**

- **Entropy, Energy, and Ergodicity.** Can long-run valuation be organized through a unified language of entropy, energy-like objects, and state ergodicity?
- **Polarization, phase, and frequency.** How can regime shifts, phase transitions, and frequency-domain structure be incorporated into asset-pricing dynamics?
- **Portfolio and projection.** Tradable portfolio's loading on co-movement latent factors and their dynamics
- **Sampling and Econometrics**

Think Deep and Represent Clear

Polish Taste and Iterate on the Path

Thank you!